



Cumulated Gain-Based Evaluation of IR Techniques

KALERVO JÄRVELIN and JAANA KEKÄLÄINEN

University of Tampere

Modern large retrieval environments tend to overwhelm their users by their large output. Since all documents are not of equal relevance to their users, highly relevant documents should be identified and ranked first for presentation. In order to develop IR techniques in this direction, it is necessary to develop evaluation approaches and methods that credit IR methods for their ability to retrieve highly relevant documents. This can be done by extending traditional evaluation methods, that is, recall and precision based on binary relevance judgments, to graded relevance judgments. Alternatively, novel measures based on graded relevance judgments may be developed. This article proposes several novel measures that compute the cumulative gain the user obtains by examining the retrieval result up to a given ranked position. The first one accumulates the relevance scores of retrieved documents along the ranked result list. The second one is similar but applies a discount factor to the relevance scores in order to devalue late-retrieved documents. The third one computes the relative-to-the-ideal performance of IR techniques, based on the cumulative gain they are able to yield. These novel measures are defined and discussed and their use is demonstrated in a case study using TREC data: sample system run results for 20 queries in TREC-7. As a relevance base we used novel graded relevance judgments on a four-point scale. The test results indicate that the proposed measures credit IR methods for their ability to retrieve highly relevant documents and allow testing of statistical significance of effectiveness differences. The graphs based on the measures also provide insight into the performance IR techniques and allow interpretation, for example, from the user point of view.

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General Terms: Experimentation, Measurement, Performance

Additional Key Words and Phrases: Graded relevance judgments, cumulated gain

1. INTRODUCTION

Modern large retrieval environments tend to overwhelm their users by their large output. Since all documents are not of equal relevance to their users, highly relevant documents, or document components, should be identified and ranked first for presentation. This is often desirable from the user point of view. In order to develop IR techniques in this direction, it is necessary to develop

This research was supported by the Academy of Finland under the grant numbers 49157 and 52894. Authors' address: Department of Information Studies, FIN-33014 University of Tampere, Finland; email: {kalervo.jarvelin, jaana.kekalainen}@uta.fi.

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evaluation approaches and methods that credit IR methods for their ability to retrieve highly relevant documents.

The current practice of liberal binary judgment of topical relevance gives equal credit for a retrieval technique for retrieving highly and marginally relevant documents. For example, TREC is based on binary relevance judgments with a very low threshold for accepting a document as relevant—the document needs to have at least one sentence pertaining to the request to count as relevant [TREC 2001]. Therefore differences between sloppy and excellent retrieval techniques, regarding highly relevant documents, may not become apparent in evaluation. To bring such differences into daylight, both graded relevance judgments and a method for using them are required.

In most laboratory tests in IR documents are judged relevant or irrelevant with regard to the request. In some studies relevance judgments are allowed to fall into more than two categories, but only a few tests actually take advantage of different relevance levels (e.g., Hersh and Hickam [1995] and Järvelin and Kekäläinen [2000]). More often relevance is conflated into two categories at the analysis phase because of the calculation of precision and recall (e.g., Blair and Maron [1985] and Saracevic et al. [1988]). However, graded relevance judgments may be collected in field studies [Vakkari and Hakala 2000; Spink et al. 1998] and also produced for laboratory test collections [Sormunen 2001; Voorhees 2001], so they are available.

Graded relevance judgments may be used for IR evaluation, first, by extending traditional evaluation measures, such as recall and precision and P–R curves, to use them. Järvelin and Kekäläinen [2000; Kekäläinen and Järvelin, 2002a] propose the use of each relevance level separately in recall and precision calculation. Thus different P–R curves are drawn for each level. They demonstrate that differing performance of IR techniques at different levels of relevance may thus be observed and analyzed. Kekäläinen and Järvelin [2002a] generalize recall and precision calculation to directly utilize graded document relevance scores. They consider precision as a function of recall, but the approach extends to DCV (Document Cut-off Value) -based recall and precision as well. They demonstrate that the relative effectiveness of IR techniques, and the statistical significance of their performance differences, may vary according to the relevance scales used.

In the present article we develop several new evaluation measures that seek to estimate the cumulative relevance gain the user receives by examining the retrieval result up to a given rank. The first one accumulates the relevance scores of retrieved documents along the ranked result list. The second one is similar but applies a discount factor to the relevance scores in order to devalue late-retrieved documents. The third one computes the relative-to-the-ideal performance of IR techniques, based on the cumulated gain they are able to yield. The first two were originally presented in Järvelin and Kekäläinen [2000] and were also applied in the TREC Web Track 2001 [Voorhees 2001] and in a text summarization experiment by Sakai and Sparck-Jones [2001]. These novel measures are akin to the average search length [briefly, ASL; Losee 1998], sliding ratio [Korfhage 1997], and normalized recall [Pollack 1968; Salton and McGill 1983; Korfhage 1997] measures. They also have

some resemblance to the ranked half-life and relative relevance measures proposed by Borlund and Ingwersen [1998] for interactive IR. However, they offer several advantages by taking both the degree of relevance¹ and the rank position (determined by the probability of relevance) of a document into account.

The novel measures are first defined and discussed and then their use is demonstrated in a case study on the effectiveness of TREC-7 runs in retrieving documents of various degrees of relevance. The results indicate that the proposed measures credit IR methods for their ability to retrieve highly relevant documents and allow testing of statistical significance of effectiveness differences. The graphs based on the measures also provide insight into the performance IR techniques and allow interpretation, for example, from the user point of view.

Section 2 explains our evaluation measures: the cumulated gain-based evaluation measures. Section 3 presents the case study. The test environment, relevance judgments, and the retrieval results are reported. Section 4 contains discussion and Section 5 conclusions.

2. CUMULATED GAIN -BASED MEASUREMENTS

2.1 Direct Cumulated Gain

When examining the ranked result list of a query, it is obvious that:

- highly relevant documents are more valuable than marginally relevant documents, and
- the greater the ranked position of a relevant document, the less valuable it is for the user, because the less likely it is that the user will ever examine the document.

The first point leads to comparison of IR techniques through test queries by their cumulated gain by document rank. In this evaluation, the relevance score of each document is somehow used as a gained value measure for its ranked position in the result and the gain is summed progressively from ranked position 1 to n . Thus the ranked document lists (of some determined length) are turned to gained value lists by replacing document IDs by their relevance scores. Assume that the relevance scores 0 to 3 are used (3 denoting high value, 0 no value). Turning document lists up to rank 200 to corresponding value lists gives vectors of 200 components each having the value 0, 1, 2, or 3. For example,

$$G' = \langle 3, 2, 3, 0, 0, 1, 2, 2, 3, 0, \dots \rangle.$$

The cumulated gain at ranked position i is computed by summing from position 1 to i when i ranges from 1 to 200. Formally, let us denote position i in the gain vector G by $G[i]$. Now the cumulated gain vector CG is defined recursively

¹For a discussion of the degree of relevance and the probability of relevance, see Robertson and Belkin [1978].

as the vector CG where:

$$CG[i] = \begin{cases} G[1], & \text{if } i = 1 \\ CG[i - 1] + G[i], & \text{otherwise.} \end{cases} \quad (1)$$

For example, from G' we obtain $CG' = \langle 3, 5, 8, 8, 8, 9, 11, 13, 16, 16, \dots \rangle$. The cumulated gain at any rank may be read directly; for example, at rank 7 it is 11.

2.2 Discounted Cumulated Gain

The second point above stated that the greater the ranked position of a relevant document, the less valuable it is for the user, because the less likely it is that the user will ever examine the document due to time, effort, and cumulated information from documents already seen. This leads to comparison of IR techniques through test queries by their cumulated gain based on document rank with a rank-based discount factor. The greater the rank, the smaller the share of the document score that is added to the cumulated gain.

A discounting function is needed that progressively reduces the document score as its rank increases but not too steeply (e.g., as division by rank) to allow for user persistence in examining further documents. A simple way of discounting with this requirement is to divide the document score by the log of its rank. For example, ${}^2\log 2 = 1$ and ${}^2\log 1024 = 10$, thus a document at the position 1024 would still get one tenth of its face value. By selecting the base of the logarithm, sharper or smoother discounts can be computed to model varying user behavior. Formally, if b denotes the base of the logarithm, the cumulated gain vector with discount DCG is defined recursively as the vector DCG where:

$$DCG[i] = \begin{cases} CG[i], & \text{if } i < b \\ DCG[i - 1] + G[i]/{}^b\log i, & \text{if } i \geq b. \end{cases} \quad (2)$$

Note that we must not apply the logarithm-based discount at rank 1 because ${}^b\log 1 = 0$. Moreover, we do not apply the discount case for ranks less than the logarithm base (it would give them a boost). This is also realistic, since the higher the base, the lower the discount and the more likely the searcher is to examine the results at least up to the base rank (say, 10).

For example, let $b=2$. From G' given in the preceding section we obtain $DCG' = \langle 3, 5, 6.89, 6.89, 6.89, 7.28, 7.99, 8.66, 9.61, 9.61, \dots \rangle$.

The (lack of) ability of a query to rank highly relevant documents toward the top of the result list should show on both the cumulated gain by document rank (CG) and the cumulated gain with discount by document rank (DCG) vectors. By averaging over a set of test queries, the average performance of a particular IR technique can be analyzed. Averaged vectors have the same length as the individual ones and each component i gives the average of the i th component in the individual vectors. The averaged vectors can be directly visualized as gain-by-rank graphs (Section 3).

To compute the averaged vectors, we need vector sum operation and vector multiplication by a constant. Let $V = \langle v_1, v_2, \dots, v_k \rangle$ and $W = \langle w_1, w_2, \dots, w_k \rangle$ be two vectors. Their sum is the vector $V + W = \langle v_1 + w_1, v_2 + w_2, \dots, v_k + w_k \rangle$. For a set of vectors $\mathbf{V} = \{V_1, V_2, \dots, V_n\}$, each of k components, the sum vector

is generalized as $\Sigma_{V \in \mathbf{V}} V = V_1 + V_2 + \dots + V_n$. The multiplication of a vector $V = \langle v_1, v_2, \dots, v_k \rangle$ by a constant r is the vector $r^*V = \langle r^*v_1, r^*v_2, \dots, r^*v_k \rangle$. The average vector $\mathbf{AV}$ based on vectors $\mathbf{V} = V_1, V_2, \dots, V_n$, is given by the function *avg-vect*($\mathbf{V}$):

$$\text{avg-vect}(\mathbf{V}) = |\mathbf{V}|^{-1} * \Sigma_{V \in \mathbf{V}} V. \quad (3)$$

Now the average CG and DCG vectors for vector sets $\mathbf{CG}$ and $\mathbf{DCG}$, over a set of test queries, are computed by *avg-vect*($\mathbf{CG}$) and *avg-vect*($\mathbf{DCG}$).

The actual CG and DCG vectors by a particular IR method may also be compared to the theoretically best possible. The latter vectors are constructed as follows. Let there be k, l , and m relevant documents at the relevance levels 1, 2, and 3 (respectively) for a given request. First fill the vector positions $1, \dots, m$ by the values 3, then the positions $m + 1, \dots, m + l$ by the values 2, then the positions $m + l + 1, \dots, m + l + k$ by the values 1, and finally the remaining positions by the values 0. More formally, the theoretically best possible score vector $\mathbf{BV}$ for a request of k, l , and m relevant documents at the relevance levels 1, 2, and 3 is constructed as follows.

$$\mathbf{BV}[i] = \begin{cases} 3, & \text{if } i \leq m, \\ 2, & \text{if } m < i \leq m + l, \\ 1, & \text{if } m + l < i \leq m + l + k, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

A sample ideal gain vector could be:

$$\mathbf{I}' = \langle 3, 3, 3, 2, 2, 2, 1, 1, 1, 1, 0, 0, 0, \dots \rangle.$$

The ideal CG and DCG vectors, as well as the average ideal CG and DCG vectors and curves, are computed as above. Note that the curves turn horizontal when no more relevant documents (of any level) can be found (Section 3 gives examples). They do not unrealistically assume as a baseline that all retrieved documents could be maximally relevant. The vertical distance between an actual (average) (D)CG curve and the theoretically best possible (average) curve shows the effort wasted on less than perfect documents due to a particular IR method. Based on the sample ideal gain vector $\mathbf{I}'$, we obtain the ideal CG and DCG ($b = 2$) vectors:

$$\mathbf{CG}'_I = \langle 3, 6, 9, 11, 13, 15, 16, 17, 18, 19, 19, 19, 19, \dots \rangle$$

$$\mathbf{DCG}'_I = \langle 3, 6, 7.89, 8.89, 9.75, 10.52, 10.88, 11.21, 11.53, 11.83, 11.83, 11.83, \dots \rangle.$$

Note that the ideal vector is based on the recall base of the search topic rather than on the result of some IR technique. This is an important difference with respect to some related measures, for example, the sliding ratio and satisfaction measure [Korfhage 1997].

2.3 Relative to the Ideal Measure—the Normalized (D)CG Measure

Are two IR techniques significantly different in effectiveness from each other when evaluated through (D)CG curves? In the case of P–R performance, we may use the average of interpolated precision figures at standard points of

operation, for example, 11 recall levels or DCV points, and then perform a statistical significance test. The practical significance may be judged by the Sparck-Jones [1974] criteria; for example, differences less than 5% are marginal and differences over 10% are essential. P–R performance is also relative to the ideal performance: 100% precision over all recall levels. The (D)CG curves are not relative to an ideal. Therefore it is difficult to assess the magnitude of the difference of two (D)CG curves and there is no obvious significance test for the difference of two (or more) IR techniques either. One needs to be constructed.

The (D)CG vectors for each IR technique can be normalized by dividing them by the corresponding ideal (D)CG vectors, component by component. In this way, for any vector position, the normalized value 1 represents ideal performance, and values in the range $[0, 1)$ the share of ideal performance cumulated by each technique. Given an (average) (D)CG vector $V = \langle v_1, v_2, \dots, v_k \rangle$ of an IR technique, and the (average) (D)CG vector $I = \langle i_1, i_2, \dots, i_k \rangle$ of ideal performance, the normalized performance vector $n(D)CG$ is obtained by the function:

$$norm\text{-}vect(V, I) = \langle v_1/i_1, v_2/i_2, \dots, v_k/i_k \rangle. \quad (5)$$

For example, based on CG' and CG'_I from above, we obtain the normalized CG vector

$$\begin{aligned} nCG' &= norm\text{-}vect(CG', CG'_I) \\ &= \langle 1, 0.83, 0.89, 0.73, 0.62, 0.6, 0.69, 0.76, 0.89, 0.84, \dots \rangle. \end{aligned}$$

The normalized DCG vector $nDCG'$ is obtained in a similar way from DCG' and DCG'_I . Note that, as a special case, the normalized ideal (D)CG vector is always $norm\text{-}vect(I, I) = \langle 1, 1, \dots, 1 \rangle$, when I is the ideal vector.

The area between the normalized ideal (D)CG vector and the normalized (D)CG vector represents the quality of the IR technique. Normalized (D)CG vectors for two or more IR techniques also have a normalized difference. These can be compared in the same way as P–R curves for IR techniques. The average of a (D)CG vector (or its normalized variation), up to a given ranked position, summarizes the vector (or performance) and is analogous to the uninterpolated average precision of a DCV curve up to the same given ranked position. The average of an $(n)(D)CG$ vector V up to the position k is given by:

$$avg\text{-}pos(V, k) = k^{-1} * \sum_{i=1 \dots k} V[i]. \quad (6)$$

These vector averages can be used in statistical significance tests in the same way as average precision over standard points of operation, for example, 11 recall levels or DCV points.

2.4 Comparison to Earlier Measures

The novel measures have several advantages when compared with several previous and related measures. The *average search length* measure [Losee 1998] estimates the average position of a relevant document in the retrieved list. The *expected search length* (ESL) measure [Korfhage 1997; Cooper 1968] is the average number of documents that must be examined to retrieve a given number of

relevant documents. Both are dichotomical; they do not take the degree of document relevance into account. The former also is heavily dependent on outliers (relevant documents found late in the ranked order).

The normalized recall measure (NR, for short; Rocchio [1966] and Salton and McGill [1983]), the sliding ratio measure (SR, for short; Pollack [1968] and Korfhage [1997]), and the satisfaction—frustration—total measure (SFT, for short; Myaeng and Korfhage [1990] and Korfhage [1997]) all seek to take into account the order in which documents are presented to the user. *The NR measure* compares the actual performance of an IR technique to the ideal one (when all relevant documents are retrieved first). Basically it measures the area between the ideal and the actual curves. NR does not take the degree of document relevance into account and is highly sensitive to the last relevant document found late in the ranked order.

The *SR measure* takes the degree of document relevance into account and actually computes the cumulated gain and normalizes this by the ideal cumulated gain for *the same retrieval result*. The result thus is quite similar to our nCG vectors. However, SR is heavily dependent on the retrieved list size: with a longer list the ideal cumulated gain may change essentially and this affects all normalized SR ratios from rank one onwards. Because our nCG is based on the recall base of the search topic, the first ranks of the ideal vector are not affected at all by extension of the evaluation to further ranks. Improving on normalized recall, SR is not dependent on outliers, but it is too sensitive to the actual retrieved set size. SR does not have the discount feature of our (n)DCG measure.

The *SFT measure* consists of three components similar to the SR measure. The satisfaction measure only considers the retrieved relevant documents, the frustration measure only the irrelevant documents, and the total measure is a weighted combination of the two. Like SR, also SFT assumes the same retrieved list of documents, which are obtained in different orders by the IR techniques to be compared. This is an unrealistic assumption for comparison since for any retrieved list size n , when $n \ll N$ (the database size), different IR techniques may retrieve quite different documents; that is the whole idea (!). A strong feature of SFT comes from its capability of punishing an IR technique for retrieving irrelevant documents while rewarding the relevant ones. SFT does not have the discount feature of our nDCG measure.

The relative relevance and ranked half-life measures [Borlund and Ingwersen 1998; Borlund 2000] were developed for interactive IR evaluation. The *relative relevance* (RR, for short) measure is based on comparing the match between the system-dependent probability of relevance and the user-assessed degree of relevance, the latter by the real person in need or a panel of assessors. The match is computed by the cosine coefficient [Borlund 2000] when *the same* ranked IR technique output is considered as vectors of relevance weights as estimated by the technique, by the user, or by the panel. RR is (intended as) an association measure between types of relevance judgments, and is not directly a performance measure. Of course, if the cosine between the IR technique scores and the user relevance judgments is low, the technique cannot perform well from the user point of view. The ranked order of documents is not taken into account.

The *ranked half-life* (RHL, for short) measure gives the median point of accumulated relevance for a given query result. It thus improves on ASL by taking the degree of document relevance into account. Like ASL, RHL is dependent on outliers. The RHL may also be the same for quite differently performing queries. RHL does not have the discount feature of DCG.

The strengths of the proposed CG, DCG, nCG, and nDCG measures can now be summarized as follows.

- They combine the degree of relevance of documents and their rank (affected by their probability of relevance) in a coherent way.
- At any number of retrieved documents examined (rank), CG and DCG give an estimate of the cumulated gain as a single measure no matter what the recall base size is.
- They are not heavily dependent on outliers (relevant documents found late in the ranked order) since they focus on the gain cumulated from the beginning of the result up to any point of interest.
- They are obvious to interpret; they are more direct than P–R curves by explicitly giving the number of documents for which each n(D)CG value holds. P–R curves do not make the number of documents explicit for given performance and may therefore mask bad performance [Losee 1998].

In addition, the DCG measure has the following further advantages.

- It realistically weights down the gain received through documents found later in the ranked results.
- It allows modeling user persistence in examining long ranked result lists by adjusting the discounting factor.

Furthermore, the normalized nCG and nDCG measures support evaluation.

- They represent performance as relative to the ideal based on a known (possibly large) recall base of graded relevance judgments.
- The performance differences between IR techniques are also normalized in relation to the ideal thereby supporting the analysis of performance differences.

Järvelin and Kekäläinen have earlier proposed recall and precision-based evaluation measures to work with graded relevance judgments [Järvelin and Kekäläinen 2000; Kekäläinen and Järvelin 2002a]. They first propose the use of each relevance level separately in recall and precision calculation. Thus different P–R curves are drawn for each level. Performance differences at different relevance levels between IR techniques may thus be analyzed. Furthermore, they generalize recall and precision calculation to directly utilize graded document relevance scores. They consider precision as a function of recall and demonstrate that the relative effectiveness of IR techniques, and the statistical significance of their performance differences, may vary according to the relevance scales used. The proposed measures are similar to standard IR measures while taking document relevance scores into account. They do not have the discount feature of our (n)DCG measure. The measures proposed in this article

are directly user-oriented in calculating the gain cumulated by consulting an explicit number of documents. P–R curves tend to hide this information. The generalized P–R approach extends to DCV-based recall and precision as well, however.

The limitations of the measures are considered in Chapter 4.

3. CASE STUDY: COMPARISON OF SOME TREC-7 RESULTS AT DIFFERENT RELEVANCE LEVELS

We demonstrate the use of the proposed measures in a case study testing runs from the TREC-7 ad hoc track with binary and nonbinary relevance judgments. We give the results as CG and DCG curves, which exploit the degrees of relevance. We further show the results as normalized nCG and nDCG curves, and present the results of a statistical test based on the averages of n(D)CG vectors.

3.1 TREC-7 Data

The seventh Text Retrieval Conference (TREC-7) had an ad hoc track in which the participants produced queries from topic statements—50 altogether—and ran those queries against the TREC text document collection. The collection included about 528,000 documents, or 1.9 GB data. Participants returned lists of the best 1000 documents retrieved for each topic. These lists were evaluated against binary relevance judgments provided by the TREC organizers (National Institute of Standards and Technology, NIST). Participants were allowed to submit up to three different runs, which could be based on different queries or different retrieval methods. [Voorhees and Harman 1999.]

The ad hoc task had two subtracks, automatic and manual, with different query construction techniques. An automatic technique means deriving a query from a topic statement without manual intervention; manual technique is anything else [Voorhees and Harman 1999].

In the case study, we used result lists for 20 topics by five participants from the TREC-7 ad hoc manual track. These topics were selected because of the availability of nonbinary relevance judgments for them (see Sormunen [2002]).²

3.2 Relevance Judgments

The nonbinary relevance judgments were obtained by rejudging documents judged relevant by NIST assessors and about 5% of irrelevant documents for each topic. The new judgments were made by six Master's students of information studies, all of them fluent in English although not native speakers. The relevant and irrelevant documents were pooled, and the judges did not know the number of documents previously judged relevant or irrelevant in the pool [Sormunen 2002].

The assumption about relevance in the rejudgment process was topicality. This agrees with the TREC judgments for the ad hoc track: documents are judged one by one; general information with limitations given in the topic's

²The numbers of topics are: 351, 353, 355, 358, 360, 362, 364, 365, 372, 373, 377, 378, 384, 385, 387, 392, 393, 396, 399, 400. For details see http://trec.nist.gov/data/topics_eng/topics.351-400.gz.

Table I. Distribution of New Relevance Judgments with Relation to Original TREC Judgments

Levels of Relevance	TREC Relevant		TREC Irrelevant		Total	
	# of Docs	%	# of Docs	%	# of Docs	%
Rel = 0	691	25.0	2780	93.8	3471	60.5
Rel = 1	1004	36.2	134	4.5	1138	19.8
Rel = 2	724	26.1	40	1.3	764	13.3
Rel = 3	353	12.7	11	0.4	364	6.4
Total	2772	100.0	2965	100.0	5737	100.0

narrative is searched, not details in the sense of question answering. New judgments were done on a four-point scale.

1. *Irrelevant document.* The document does not contain any information about the topic.
2. *Marginally relevant document.* The document only points to the topic. It does not contain more or other information than the topic statement.
3. *Fairly relevant document.* The document contains more information than the topic statement but the presentation is not exhaustive. In the case of a multifaceted topic, only some of the subthemes are covered.
4. *Highly relevant document.* The document discusses the themes of the topic exhaustively. In the case of multifaceted topics, all or most subthemes are covered.

Altogether 20 topics from TREC-7 and 18 topics from TREC-8 were reassessed. In Table I the results of rejudgment are shown with respect to the original TREC judgments. It is obvious that almost all originally irrelevant documents were also assessed irrelevant in rejudgment (93.8%). Of the TREC relevant documents 75% were judged relevant at some level, and 25% irrelevant. This seems to indicate that the reassessors have been somewhat stricter than the original judges. The great overlap in irrelevant documents proves the new judgments reliable. However, in the case study we are not interested in comparing the results based on different judgments but in showing the effects of utilizing nonbinary relevance judgments in evaluation. Thus we do not use the original TREC judgments in any phase of the case study.

In the subset of 20 topics, among all relevant documents ($N = 1182$), the share of highly relevant documents was 20.1%, the share of fairly relevant documents was 30.5%, and that of marginal documents was 49.4%.

3.3 The Application of the Evaluation Measures

We ran the TREC-7 result lists of five participating groups against the new, graded relevance judgments. For the cumulated gain evaluations we tested logarithm bases and handling of relevance levels varied as parameters as follows.

1. We tested different relevance weights at different relevance levels. First, we replaced document relevance levels 0, 1, 2, 3 with binary weights; that is, we gave documents at level 0 weight 0, and documents at levels 1 to 3 weight 1 (weighting scheme 0–1–1–1 for the four-point scale). Then we

replaced the relevance levels with weights 0, 0, 0, 1, to test the other extreme where only the highly relevant documents are valued. The last weighting scheme, 0, 1, 10, 100, is between the extremes; the highly relevant documents are valued one hundred times more than marginally relevant documents, and ten times more than fairly relevant ones. Different weighting on highly relevant documents may affect the relative effectiveness of IR techniques as also pointed out by Voorhees [2001]. The first and last weighting schemes only are shown in graphs because the 0–0–0–1 scheme is very similar to the last one in appearance.

2. The logarithm bases 2 and 10 were tested for the DCG vectors. The base-2 models impatient users and base-10 persistent ones. Although the differences in results do not vary markedly with the logarithm base, we show only the results for the logarithm base-2. We also prefer the stricter test condition the smaller logarithm base provides.
3. The average actual CG and DCG vectors were compared to the ideal average vectors.
4. The average actual CG and DCG vectors were normalized by dividing by with the ideal average vectors.

3.4 Cumulated Gain

Figures 1(a) and (b) present the CG vector curves for the five runs at ranks 1 to 100, and the ideal curves. Figure 1(a) shows the weighting scheme 0–1–1–1, and 1(b) the scheme 0–1–10–100. In the ranked result list, highly relevant documents add either 1 or 100 points to the cumulated gain; fairly relevant documents add either 1 or 10 points; marginally relevant documents add 1 point; and irrelevant documents add 0 points to the gain.

The different weighting schemes change the position of the curves compared to each other. For example, in Figure 1(a) (the binary weighting scheme) the performance of (run) D is close to that of C; when highly relevant documents are given more weight, D is more similar to B, and C and E are close in performance. Note that the graphs have different scales because of the weighting schemes.

In Figure 1(a) the best possible curve starts to level off at rank 100 reflecting the fact that at rank 100 practically all relevant documents have been found. In Figure 1(b) it can be observed that the ideal curve has already found the most fairly and highly relevant documents by the rank 50. This, of course, reflects the sizes of the recall bases: the average number of documents at relevance levels 2 and 3 per topic is 29.9. The best system (E) hangs below the ideal by 0–39 points with binary weights (1(a)), and 70 to 894 points with nonbinary weights (1(b)). Note that the differences are not greatest at rank 100 but often earlier. The other runs remain further below by 0 to 6 points with binary weights (1(a)), and 0 to 197 points with nonbinary weights (1(b)). The differences between the ideal and all actual curves are all bound to diminish when the ideal curve levels off.

The curves can also be interpreted in another way. In Figure 1(a) one has to retrieve 30 documents by the best run, and 90 by the worst run in order to gain the benefit that could theoretically be gained by retrieving only 10 documents (the ideal curve). In this respect the best run is three times as effective as the

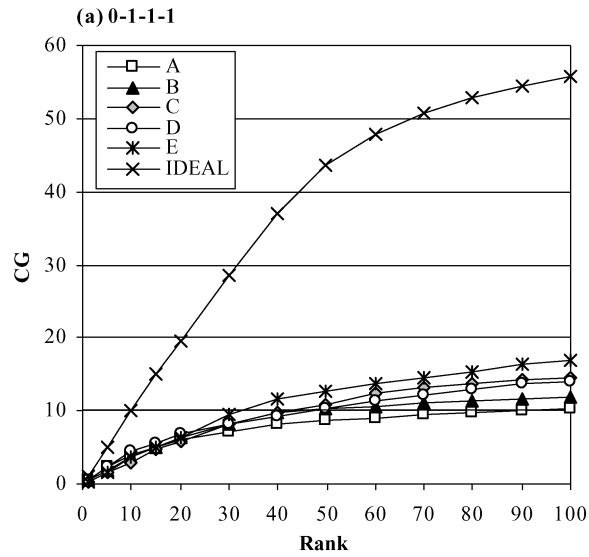


Fig. 1(a). Cumulated gain (CG) curves, binary weighting.

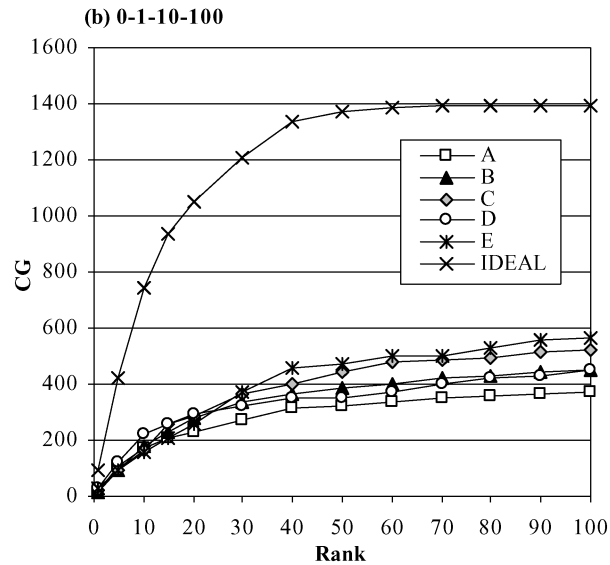


Fig. 1(b). Cumulated gain (CG) curves, nonbinary weighting.

worst run. In Figure 1(b) one has to retrieve 35 documents by the best run to get the benefit theoretically obtainable at rank 5; the worst run does not provide the same benefit even at rank 100.

3.5 Discounted Cumulated Gain

Figures 2(a) and (b) show the DCG vector curves for the five runs at ranks 1 to 100, and the ideal curve. The $\log_2$ of the document rank is used as the

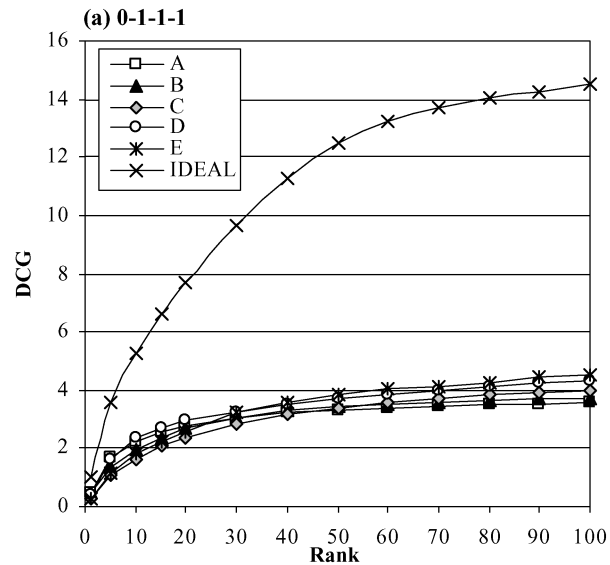


Fig. 2(a). Discounted cumulated gain (DCG) curves, binary weighting.

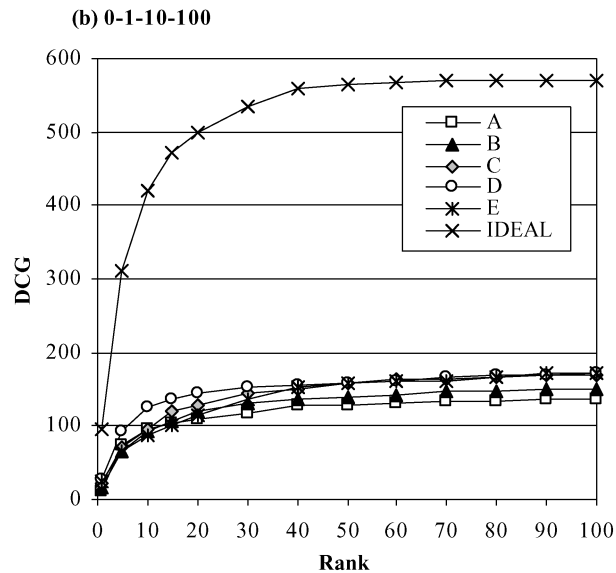


Fig. 2(b). Discounted cumulated gain (DCG) curves, nonbinary weighting.

discounting factor. Discounting alone seems to narrow the differences between the systems (1(a) compared to 2(a) and 1(b) to 2(b)). Discounting and nonbinary weighting changes the performance order of the systems: in Figure 2(b), run A seems to lose and run C to benefit.

In Figure 2(a), the ideal curve levels off upon the rank 100. The best run hangs below by 0.5 to 10 points. The other runs remain further below by 0.25 to

1 point. Thus, with the discounting factor and binary weighting, the runs seem to perform equally. In Figure 2(b), the ideal curve levels off upon the rank 50. The best run hangs below by 71 to 408 points. The other runs remain further below by 13 to 40 points. All the actual curves still grow at the rank 100, but beyond that the differences between the best possible and the other curves gradually become stable.

These graphs can also be interpreted in another way: In Figure 2(a) one has to expect the user to examine 40 documents by the best run in order to gain the (discounted) benefit that could theoretically be gained by retrieving only 5 documents. The worst run reaches that gain around rank 95. In Figure 2(b), none of the runs gives the gain that would theoretically be obtainable at rank 5. Given the worst run, the user has to examine 50 documents in order to get the (discounted) benefit that is obtained with the best run at rank 10. In that respect the difference in the effectiveness of runs is essential.

One might argue that if the user goes down to, say, 50 documents, she gets the real value, not the discounted one, and therefore the DCG data should not be used for effectiveness comparison. Although this may hold for the user situation, the DCG-based comparison is valuable for the system designer. The user is less and less likely to scan further and thus documents placed there do not have their real relevance value, a retrieval technique placing relevant documents later in the ranked results should not be credited as much as another technique ranking them earlier.

3.6 Normalized (D)CG Vectors and Statistical Testing

Figures 3(a) and (b) show the curves for CG vectors normalized by the ideal vectors. The curve for the normalized ideal CG vector has value 1 at all ranks. The actual normalized CG vectors reach it in due course when all relevant documents have been found. Differences at early ranks are easier to observe than in Figure 1. The nCG curves readily show the differences between methods to be compared because of the same scale but they lack the straightforward interpretation of the gain at each rank given by CG curves. In Figure 3(b) the curves start lower than in Figure 3(a); it is obvious that highly relevant documents are more difficult to retrieve.

Figures 4(a) and (b) display the normalized curves for DCG vectors. The curve for the normalized ideal DCG vector has value 1 at all ranks. The actual normalized DCG vectors never reach it; they start to level off upon rank 100. The effect of discounting can be seen by comparing Figures 3 and 4; for example, the order of the runs changes. The effect of normalization can be detected by comparing Figures 2 and 4: the differences between the IR techniques are easier to detect and comparable.

Statistical testing of differences between query types was based on normalized average n(D)CG vectors. These vector averages can be used in statistical significance tests in the same way as average precision over document cutoff values. The classification we used to label the relevance levels through numbers 0 to 3 is on an ordinal scale. Holding to the ordinal scale suggests nonparametric statistical tests, such as the Friedman test (see Conover [1980]). However,

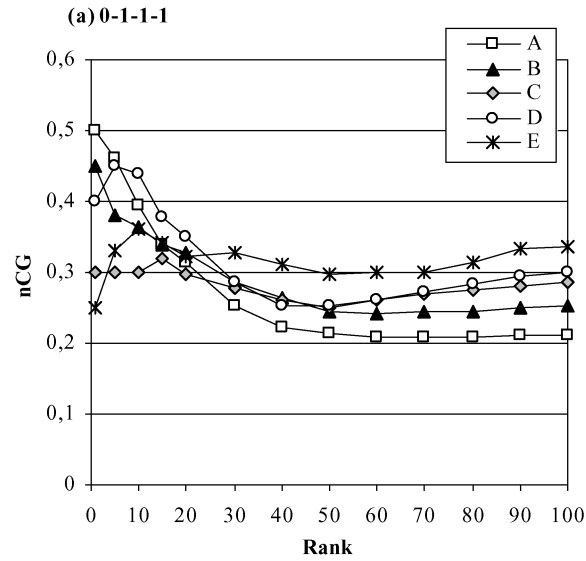


Fig. 3(a). Normalized cumulated gain (nCG) curves, binary weighting.

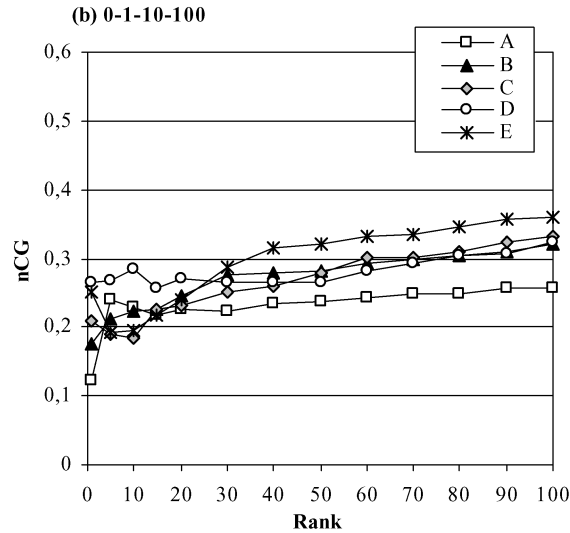


Fig. 3(b). Normalized cumulated gain (nCG) curves, nonbinary weighting.

we have based our calculations on class weights to represent their relative differences. The weights 0, 1, 10, and 100 denote differences on a ratio scale. This suggests the use of parametric tests such as ANOVA provided that its assumptions on sampling and measurement distributions are met. Next we give the grand averages of the vectors of length 200, and the results of the Friedman test; ANOVA did not prove any differences significant.

In Table II, the average is first calculated for each topic, then an average is taken over the topics. If the average had been taken of vectors of different

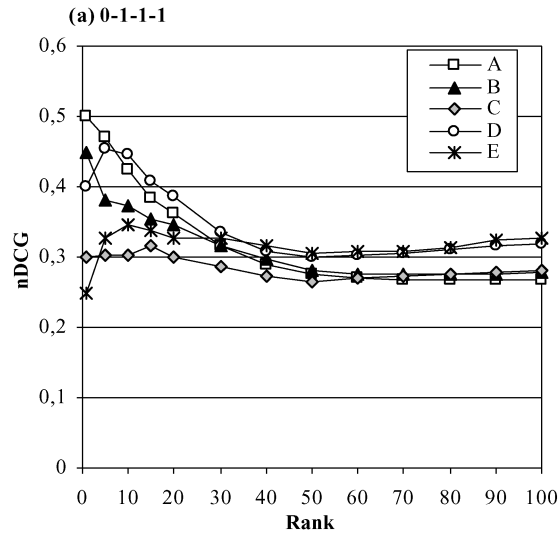


Fig. 4(a). Normalized discounted cumulated gain (nDCG) curves, binary weighting.

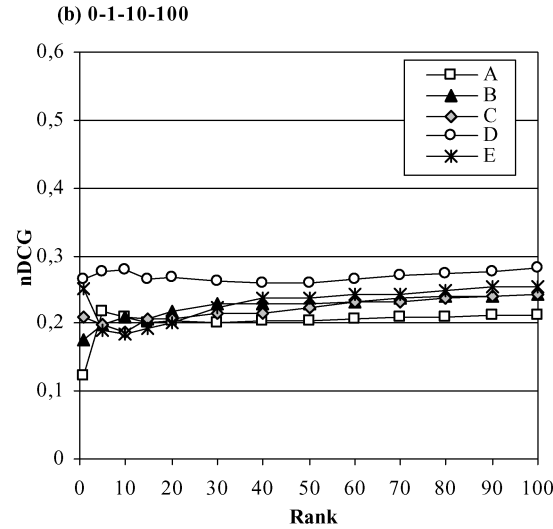


Fig. 4(b). Normalized discounted cumulated gain (nDCG) curves, nonbinary weighting.

lengths, the results of the statistical tests might have changed. Also, the number of topics (20) is rather small to provide reliable results. However, even these data illuminate the behavior of the (n)(D)CG measures.

4. DISCUSSION

The proposed measures are based on several parameters: the last rank considered, the gain values to employ, and discounting factors to apply. An experimenter needs to know which parameter values and combinations to use. In practice, the evaluation context and scenario should suggest these values.

Table II. n(D)CG Averages over Topics and Statistical Significance the Results for Five TREC-7 runs^a

	A	B	C	D	E	Stat. Sign.
nCG (0-1-1-1)	0.242	0.271	0.293	0.318	0.343	
nDCG (0-1-1-1)	0.292	0.294	0.287	0.335	0.331	
nCG (0-1-10-100)	0.254	0.305	0.313	0.316	0.342	D, E > A**
nDCG (0-1-10-100)	0.211	0.238	0.236	0.279	0.247	D > A*
nCG (0-0-0-1)	0.244	0.301	0.309	0.309	0.329	
nDCG (0-0-0-1)	0.192	0.220	0.223	0.259	0.224	

^a ** = $p < 0.01$; * = $p < 0.05$, Friedman test.

Alternatively, several values and/or combinations may be used to obtain a richer picture of IR system effectiveness under different conditions. Below we consider the effects of the parameters. Thereafter we discuss statistical testing, relevance judgments, and limitations of the measures.

Last Rank Considered

Gain vectors of various lengths from 1 to n may be used for computing the proposed measures and curves. If one analyzes the curves alone, the last rank does not matter. Eventual differences between the IR methods are observable for any rank region. The gain difference for any point (or region) of the curves may be measured directly.

If one is interested in differences in average gain up to a given last rank, then the last rank matters, particularly for nCG measurements. Suppose IR method A is somewhat better than the method B in early ranks (say, down to rank 10) but beyond them methods B starts catching up so that they are *en par* at rank 50 with all relevant documents found. If one now evaluates the methods by nCG, they might be statistically significantly different for the ranks 1 to 10, but there probably would be no significant difference for the ranks 1 to 100 (or down to lower positions).

If one uses nDCG in the previous case the difference earned by method A would be preserved due to discounting low-ranked relevant documents. In this case the difference between the methods may be statistically significant also for the ranks 1 to 100 (or down to lower positions).

The measures themselves cannot tell how they should be applied: down to which rank? This depends on the evaluation scenario and the sizes of recall bases. It makes sense to produce the n(D)CG curves liberally, that is, down to quite low ranks. The significance of differences between IR methods, when present, can be tested for selected regions (top n) when justified by the scenario. Also our test data demonstrate that one run may be significantly better than another, if just top ranks are considered, while being similarly effective as another, if low ranks are included also (say, up to 100; see, e.g., runs C and D in Figure 3).

Gain Values

Justifying different gain values for documents relevant to different degrees is inherently quite arbitrary. It is often easy to say that one document is more

relevant than another, but the quantification of this difference still remains arbitrary. However, determining such documents as equally relevant is another arbitrary decision, and less justified in the light of the evidence from relevance studies [Tang et al. 1999; Sormunen 2002].

Since graded relevance judgments can be provided reliably, the sensitivity of the evaluation results to different gain quantifications can easily be tested. Sensitivity testing is also typical in cost-benefit studies, so this is no new idea. Even if the evaluation scenario did not advise us on the gain quantifications, evaluation through several flat to steep quantifications informs us of the relative performance of IR methods better than a single one. Voorhees [2001] used this approach in the TREC Web Track evaluation, when she weighted highly relevant documents by factors 1 to 1000 in relation to marginal documents. Varying weighting affected the relative effectiveness order of IR systems in her test. Our present illustrative findings based on TREC data also show that weighting affects the relative effectiveness order of IR systems. We can observe in Figures 4(a) and (b) (Section 3.6) that by changing from weighting 0–1–1–1, that is, flat TREC-type weights, to weights 0–1–10–100 for the irrelevant to highly relevant documents, run D appears more effective than the others.

Tang et al. [1999] proposed seven as the optimal number of relevance levels in relevance judgments. Although our findings are for four levels, the proposed measures are not tightly coupled with any particular number of levels.

Discounting Factor

The choice between (n)CG and (n)DCG measures in evaluation is essential: discounting the gain of documents retrieved late affects the order of effectiveness of runs as we saw in Sections 3.4 and 3.5 (Figures 1(b) and 2(b)). It is, however, again somewhat arbitrary to apply any specific form of discounting. Consider the discounting case of the DCG function:

$$\text{DCG}[i] = \text{DCG}[i - 1] + G[i]/df,$$

where df is the discounting factor and i the current ranked position. There are three cases of interest.

- If $df = 1$ then $\text{DCG} = \text{CG}$ and no discounting is performed—all documents, at whatever rank retrieved, retain their relevance score.
- If $df = i$ then we have a very sharp discount—only the first documents would really matter, which is hardly desirable and realistic for evaluation.
- If $df = b^{\log i}$ then we have a smooth discounting factor, the smoothness of which can be adjusted by the choice of the base b . A relatively small base ($b = 2$) models an impatient searcher for whom the value of late documents drops rapidly. A relatively high base ($b > 10$) models a patient searcher for whom even late documents are valuable. A very high base ($b > 100$) yields a very marginal discount from the practical IR evaluation point of view.

We propose the use of the logarithmic discounting factor. However, the choice of the base is again somewhat arbitrary. Either the evaluation scenario should

advise the evaluator of the base or a range of bases could be tried. Note that in the DCG function case $DCG[i] = DCG[i - 1] + G[i]/b^{\log i}$, the choice of base would not affect the order of effectiveness of IR methods because $b^{\log i} = b^{\log a} \cdot a^{\log i}$ for any pair of bases a and b since $b^{\log a}$ is a constant. This is the reason for applying the discounting case for DCG only after the rank indicated by the logarithm base. This is also the point where discounting begins because $b^{\log b} = 1$. In the rank region 2 to b discounting would be replaced by boosting.

There are two borderline cases for the logarithm base. When the base b ($b \geq 1$) approaches 1, discounting becomes very aggressive and finally only the first document would matter—hardly realistic. On the other hand, if b approaches infinity, then DCG approaches CG—neither realistic. We believe that the base range 2 to 10 serves most evaluation scenarios well.

Practical Methodological Problems

The discussion above leaves open the proper parameter combinations to use in evaluation. This is unfortunate but also unavoidable. The mathematics work for whatever parameter combinations and cannot advise us on which to choose. Such advice must come from the evaluation context in the form of realistic evaluation scenarios. In research campaigns such as TREC, the scenario(s) should be selected.

If one is evaluating IR methods for very busy users who are only willing to examine a few best answers for their queries, it makes sense to evaluate down to shallow ranks only (say, 30), and use fairly sharp gain quantifications (say, 0–1–10–100) and a low base for the discounting factor (say, 2). On the other hand, if one is evaluating IR methods for patient users who are willing to dig down in the low-ranked and marginal answers for their queries, it makes sense to evaluate down to deep ranks (say, 200), and use moderate gain quantifications (say, 0–1–2–3) and a high base for the discounting factor (say, 10). It makes sense to try out both scenarios in order to see whether some IR methods are superior in one scenario only.

When such scenarios are argued out, they can be critically assessed and defended for the choices involved. If this is not done, an arbitrary choice is committed, perhaps unconsciously. For example, precision averages over 11 recall points with binary relevance gains models well only very patient users willing to dig deep down the low-ranked answers, no matter how relevant versus marginal the answers are. Clearly this is not the only scenario at which one should look.

The normalized measures nCG and nDCG we propose are normalized by the best possible behavior for each query on a rank-by-rank basis. Therefore the averages of the normalized vectors are also less prone to the problems of recall base size variation which plague the precision–recall measurements, whether they are based on DCVs or precision as a function of recall.

The cumulated gain curves illustrate the value the user actually gets, but discounted cumulated gain curves can be used to forecast system performance with regard to a user's patience in examining the result list. For example, if

the CG and DCG curves are analyzed horizontally in the case study, we may conclude that a system designer would have to expect the users to examine 100 to 500% more documents by the worst query types to collect the same gain collected by the best query types. Although it is possible that persistent users go way down the result list (e.g., from 30 to 60 documents), it often is unlikely to happen, and a system requiring such behavior is, in practice, much worse than a system yielding the gain within 50% of the documents.

Relevance Judgments

Kekäläinen and Järvelin [2002a] argue on the basis of several theoretical, laboratory, and field studies that the degree of document relevance varies and document users can distinguish among them. Some documents are far more relevant than others. Furthermore, in many studies on information seeking and retrieval, multiple degree relevance scales have been found pertinent, whereas the number of degrees employed varies. It is difficult to determine how many degrees there should be in general. This depends on the study setting and the user scenarios. When multiple degree approaches are justified, evaluation methods should utilize/support them.

TREC has been based on binary relevance judgments with a very low threshold for accepting a document as relevant for a topical request: the document needs to have at least one sentence pertaining to the request to count as relevant [TREC 2001]. This is a very special evaluation scenario and there are obvious alternatives. In many scenarios, at that level of contribution one would count the document at most as marginal unless the request is factual, in which case a short factual response should be regarded as highly relevant and another not giving the facts as marginal if not irrelevant. This is completely compatible with the proposed measures. If the share of marginal documents were high in the test collection, then utilizing TREC-like liberal binary relevance judgments would lead to difficulties in identifying the better techniques as such. In our data sample, about 50% of the relevant documents were marginally relevant. Possible differences between IR techniques in retrieving highly relevant documents might be evened up by their possible indifference in retrieving marginal documents. The net differences might seem practically marginal and statistically insignificant.

Statistical Testing

Holding to the ordinal scale of relevance judgments suggests nonparametric statistical tests, such as the Wilcoxon or Friedman tests. However, when weights are used, the scale of measurement becomes one of interval or ratio scale. This suggests the use of parametric tests such as ANOVA or *t*-test provided that their assumptions on sampling and measurement distributions are met. For example, Zobel [1998] used parametric tests when analyzing the reliability of IR experiment results. Also Hull [1993] argued that with sufficient data parametric tests may be used. In our test case ANOVA gave a result different from Friedman, an effect of the magnitude of the differences between the IR

runs considered. However, the data set used in the demonstration was fairly small.

Empirical Findings Based on the Proposed Measures

The DCG measure has been applied in the TREC Web Track 2001 [Voorhees 2001] and in a text summarization experiment by Sakai and Sparck-Jones [2001]. Voorhees' findings are based on a three-point relevance scale. She examined the effect of incorporating highly relevant documents (HRDs) into IR system evaluation and weighting them more or less sharply in a DCG-based evaluation. She found out that the relative effectiveness of IR systems is affected when evaluated by HRDs. Voorhees pointed out that moderately sharp weighting of HRDs in DCG measurement supports evaluation for HRDs but avoids problems caused by instability due to small recall bases of HRDs in test collections. Sakai and Sparck-Jones first assigned the weight 2 to each highly relevant document, and the weight 1 to each partially relevant document. They also experimented with other valuations, for example, zero for the partially relevant documents. Sakai and Sparck-Jones used log base 2 as the discounting factor to model user's (lack of) persistence. The DCG measure served to test the hypotheses in the summarization study. Our present demonstrative findings based on TREC-7 data also show that weighting affects the relative effectiveness order of IR systems. These results exemplify the usability of the cumulated gain-based approach to IR evaluation.

Limitations

The measures considered in this article, both the old and the new ones, have weaknesses in several areas. First, none of them take into account order effects on relevance judgments, or document overlap (or redundancy). In the TREC interactive track [Over 1999], *instance recall* is employed to handle this. The user-system pairs are rewarded for retrieving distinct instances of answers rather than multiple overlapping documents. In principle, the $n(D)CG$ measures may be used for such evaluation. Second, the measures considered in Section 2.4 all deal with relevance as a single dimension although it really is multidimensional [Vakkari and Hakala 2000]. In principle, such multidimensionality may be accounted for in the construction of recall bases for search topics but leads to complexity in the recall bases and in the evaluation measures. Nevertheless, such added complexity may be worth pursuing because so much effort is invested in IR evaluation.

Third, any measure based on static relevance judgments is unable to handle dynamic changes in real relevance judgments. However, when changes in user's relevance criteria lead to a reformulated query, an IR system should retrieve the best documents for the reformulated query. Kekäläinen and Järvelin [2002b] argue that complex dynamic interaction is a sequence of simple topical interactions and thus good one-shot performance by a retrieval system should be rewarded in evaluation. Changes in the user's information need and relevance criteria affect consequent requests and queries. Although this is likely to happen, it has not been shown that this should affect the design of the retrieval

techniques. Neither has it been shown that this would invalidate the proposed or the traditional evaluation measures.

It may be argued that IR systems should not rank just highly relevant documents in top ranks. Consequently, they should not be rewarded in evaluation for doing so. Spink et al. [1998] have argued that partially relevant documents are important for users at the early stages of their information-seeking process. Therefore one might require that IR systems be rewarded for retrieving partially relevant documents in the top ranks.

For about 40 years IR systems have been compared on the basis of their ability to provide relevant—or useful—documents for their users. To us it seems plausible that highly relevant documents are those which people find useful. The findings by Spink et al. do not really disqualify this belief; they rather state that students in the early states of their information seeking tend to change their relevance criteria and problem definition and that the number of partially relevant documents correlates with these changes.

However, if it should turn out that for some purposes, IR systems should rank partially relevant documents higher than, say, highly relevant documents, our measures perfectly suit comparisons on such a basis: the documents should just be weighted accordingly. We do not intend to say how or on what criteria the relevance judgments should be made; we only propose measures that take into account differences in relevance.

The limitations of the proposed measures being similar to those of traditional measures, the proposed measures offer benefits taking the degree of document relevance into account and modeling user persistence.

5. CONCLUSIONS

We have argued that in modern large database environments, the development and evaluation of IR methods should be based on their ability to retrieve highly relevant documents. This is often desirable from the user viewpoint and presents a not too liberal test for IR techniques.

We then developed novel methods for IR technique evaluation, which aim at taking the document relevance degrees into account. These are the CG and the DCG measures, which give the (discounted) cumulated gain up to any given document rank in the retrieval results, and their normalized variants nCG and nDCG, based on the ideal retrieval performance. They are related to some traditional measures such as *average search length* [Losee 1998], *expected search length* [Cooper 1968], *normalized recall* [Rocchio 1966; Salton and McGill 1983], *sliding ratio* [Pollack 1968; Korfhage 1997], *satisfaction—frustration—total measure* [Myaeng and Korfhage 1990], and *ranked half-life* [Borlund and Ingwersen 1998].

The benefits of the proposed novel measures are many. They systematically combine document rank and degree of relevance. At any number of retrieved documents examined (rank), CG and DCG give an estimate of the cumulated gain as a single measure no matter what the recall base size is. Performance is determined on the basis of recall bases for search topics and thus does not vary in an uncontrollable way, which is true of measures based on the retrieved

lists only. The novel measures are not heavily dependent on outliers since they focus on the gain cumulated from the beginning of the result up to any point of interest. They are obvious to interpret, and do not mask bad performance. They are directly user-oriented in calculating the gain cumulated by consulting an explicit number of documents. P–R curves tend to hide this information. In addition, the DCG measure realistically downweights the gain received through documents found later in the ranked results and allows modeling user persistence in examining long ranked result lists by adjusting the discounting factor. Furthermore, the normalized nCG and nDCG measures support evaluation by representing performance as relative to the ideal based on a known (possibly large) recall base of graded relevance judgments. The performance differences between IR techniques are also normalized in relation to the ideal thereby supporting the analysis of performance differences.

An essential feature of the proposed measures is the weighting of documents at different levels of relevance. What is the value of a highly relevant document compared to the value of fairly and marginally relevant documents? There can be no absolute value because this is a subjective matter that also depends on the information-seeking situation. It may be difficult to justify any particular weighting scheme. If the evaluation scenario does not suggest otherwise, several weight values may be used to obtain a richer picture of IR system effectiveness under different conditions. Regarding all at least somewhat relevant documents as equally relevant is also an arbitrary (albeit traditional) decision, and also counterintuitive.

It may be argued that IR systems should not rank just highly relevant documents in top ranks. One might require that IR systems be rewarded for retrieving partially relevant documents in the top ranks. However, our measures perfectly suit comparisons on such a basis: the documents should just be weighted accordingly. The traditional measures do not allow this.

The CG and DCG measures complement P–R based measures [Järvelin and Kekäläinen 2000; Kekäläinen and Järvelin 2002a]. Precision over fixed recall levels hides the user's effort up to a given recall level. The DCV-based precision–recall graphs are better but still do not make the value gained by ranked position explicit. The CG and DCG graphs provide this directly. The distance to the theoretically best possible curve shows the effort wasted on less than perfect or useless documents. The normalized CG and DCG graphs show explicitly the share of ideal performance given by an IR technique and make statistical comparisons possible. The advantage of the P–R based measures is that they treat requests with a different number of relevant documents equally, and from the system's point of view the precision at each recall level is comparable. In contrast, CG and DCG curves show the user's point of view as the number of documents needed to achieve a certain gain. Together with the theoretically best possible curve they also provide a stopping rule; that is, when the best possible curve turns horizontal, there is nothing to be gained by retrieving or examining further documents.

Generally, the proposed evaluation measures and the case further demonstrate that graded relevance judgments are applicable in IR experiments. The dichotomous and liberal relevance judgments generally applied may be too

permissive, and, consequently, too easily give credit to IR system performance. We believe that, in modern large environments, the proposed novel measures should be used whenever possible, because they provide richer information for evaluation.

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